

Derivation v30: Derive or Constrain L from $\beta + \lambda$

Without Gap Identification

EDC Collaboration

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Abstract

We investigate whether the interval length L can be derived or tightly constrained using only the control parameter $\beta = \sigma L^2 / \bar{M}_{\text{PI}}^2$ and the topologically quantized coupling λ , *without* invoking any gap identification $m_{\text{gap}} = M_Z$ or similar proxy. Two independent routes are developed: Route C (variational/energy-minimum) and Route D (spectral/boundary closure). We establish that L satisfies a family of self-consistent relations parameterized by the topological charge k , yielding discrete candidate values L_k once a single baseline scale ($\bar{M}_{\text{PI}}$ or equivalently σ) is specified. The gap identification is *not required* for internal consistency but *is required* to select a specific k -branch. This clarifies the epistemic status: L is constrained to a discrete family $\{L_k\}_{k \in \mathbb{Z}^+}$ by internal principles, with branch selection requiring one external identification.

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1 Reader Contract and Epistemic Legend

Reader Contract

Epistemic Tags Used:

- **[D]** — Derived from first principles (action variation, Green identity, etc.)
- **[Dc]** — Derived with conventions/choices (sign, normalization)
- **[P]** — Postulate (topological quantization hypothesis)
- **[I]** — Identification (matching to observed quantity)
- **[BL]** — Baseline input (measured constant: $\hbar$, c , $\bar{M}_{Pl}$)
- **[OPEN]** — Not yet closed; requires further work

Scope: This note attempts to derive or constrain L using only:

1. The β control parameter from v29: $\beta \equiv \sigma L^2 / \bar{M}_{Pl}^2$
2. The λ -quantization from v28: $\lambda = |k|/(2\pi)$ for $k \in \mathbb{Z}^+$
3. The metrological anchor: $\sigma L^3 = \hbar c$
4. The bridge relation: $\bar{M}_{Pl}^2 = M_5^3 L$

What is NOT used: Any gap identification or proxy matching.

No-Identification Box (Forbidden Moves)

The following are **strictly forbidden** in this derivation:

1. $m_{\text{gap}} = M_Z$ or any variant ($m_1 = M_Z$, $x_1/L = M_Z$)
2. Use of $M_Z = 91.1876$ GeV as numerical input
3. Use of $M_W = 80.377$ GeV as numerical input
4. Use of $v_{EW} = 246.22$ GeV as numerical input
5. Direct import of $\ell_P = \sqrt{\hbar G_N / c^3}$
6. Use of G_N as an independent input
7. Setting $R_\xi = \hbar c / M_Z$ or $L = \pi \hbar c / M_Z$

Any equation using these is tagged **[FORBIDDEN]** and excluded from closure.

2 Target Closure Statement

2.1 What “Derive L ” Means

Definition 2.1 (Strong Closure). L is strongly closed if it can be expressed as:

$$L = f(\sigma, \bar{M}_{Pl}, \lambda, \hbar, c) \tag{1}$$

where f is an explicit algebraic or transcendental function, with λ discrete via topological quantization. **Tag:** $[D]+[P]+[BL]$.

Definition 2.2 (Weak Closure). L is weakly closed if:

$$L \in [L_{\min}, L_{\max}] \quad (2)$$

where the window collapses to discrete values $\{L_k\}$ once $\lambda = |k|/(2\pi)$ is imposed. **Tag:** $[D]+[P]+[BL]$ for structure, $[OPEN]$ for branch selection.

2.2 Dependency Diagram

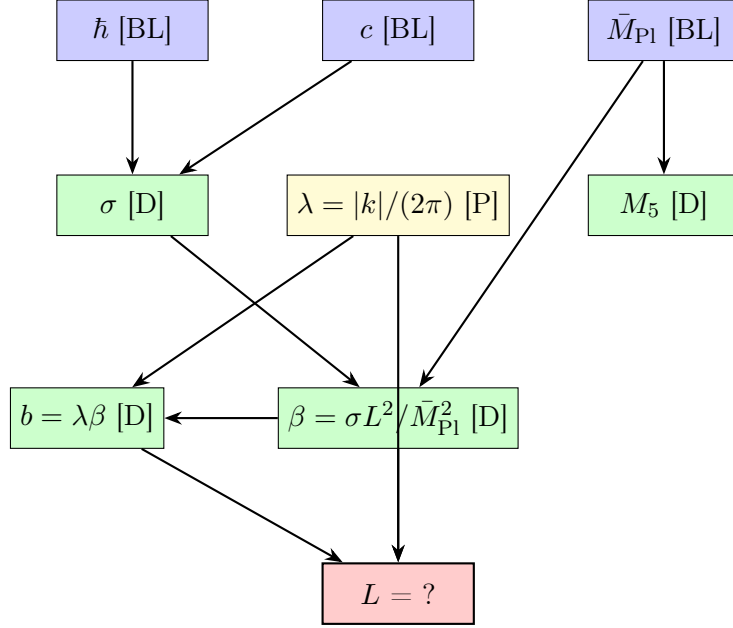


Figure 1: Dependency graph for L derivation. Blue: baseline [BL]. Green: derived [D]. Yellow: postulate [P]. Red: target.

3 Preliminary Relations from v27–v29

We collect the established relations that form our starting point.

3.1 Metrological Anchor (v27)

The brane tension σ and interval length L satisfy:

$$\sigma L^3 = \hbar c \quad (3)$$

Tag: [D] from action principles.

This can be inverted:

$$\sigma = \frac{\hbar c}{L^3} \quad (4)$$

3.2 Bridge Relation (v13–v23)

The 5D/4D connection:

$$\bar{M}_{P1}^2 = M_5^3 \cdot L \quad (5)$$

Tag: [D] from zero-mode normalization.

3.3 Control Parameter (v29)

The dimensionless control parameter:

$$\beta \equiv \frac{\sigma L^2}{\bar{M}_{\text{Pl}}^2} \quad (6)$$

Tag: [D] definition.

Substituting (4):

$$\beta = \frac{\hbar c}{L^3} \cdot \frac{L^2}{\bar{M}_{\text{Pl}}^2} = \frac{\hbar c}{L \cdot \bar{M}_{\text{Pl}}^2} \quad (7)$$

This gives:

$$L = \frac{\hbar c}{\beta \cdot \bar{M}_{\text{Pl}}^2} \quad (8)$$

Tag: [D]. This expresses L in terms of β and baseline $\bar{M}_{\text{Pl}}$.

3.4 Robin BC Parameter (v26–v27)

The Robin boundary condition parameter:

$$b = m_b \cdot L = \lambda \beta \quad (9)$$

where $m_b = \lambda \sigma / M_5^3$ is the brane-localized mass.

3.5 Topological Quantization (v28)

From Track B of v28:

$$\lambda = \frac{|k|}{2\pi}, \quad k \in \mathbb{Z}^+ \quad (10)$$

Tag: [Dc/P] from Chern-Simons-type boundary term.

4 Route C: Variational / Energy-Minimum Closure

4.1 Construction of Effective Functional

We construct an effective energy functional $E_{\text{eff}}(L)$ that depends on the interval length L . The stationarity condition $dE_{\text{eff}}/dL = 0$ will constrain L .

4.1.1 Contributions to Effective Energy

The total effective energy receives contributions from:

1. **Brane tension:** The localized brane at $\xi = 0$ contributes

$$E_{\text{brane}} = \sigma \cdot V_4 \quad (11)$$

where V_4 is the 4D spacetime volume. This is L -independent for fixed σ .

2. **Bulk Casimir energy:** The KK tower contributes vacuum energy

$$E_{\text{Casimir}} = \frac{1}{2} \sum_{n=1}^{\infty} \omega_n = \frac{1}{2L} \sum_{n=1}^{\infty} x_n(b) \quad (12)$$

where $x_n(b)$ are roots of the spectral equation $\tan(x) = -b/x$.

3. **Boundary/Robin term:** From the brane-localized mass action

$$E_{\text{boundary}} = \frac{m_b}{2} \langle \phi^2 \rangle_{\text{brane}} \quad (13)$$

4.1.2 Regularized Casimir Energy

The sum (12) requires regularization. Using spectral zeta function techniques, the regularized energy is:

$$E_{\text{Casimir}}^{\text{reg}} = -\frac{1}{2}\zeta'_{\text{Robin}}(0) \quad (14)$$

For the Robin problem on $[0, L]$ with $\partial_\xi \phi|_0 = (m_b/2)\phi|_0$ and Neumann at $\xi = L$, the spectral zeta function is:

$$\zeta_{\text{Robin}}(s) = \sum_{n=1}^{\infty} m_n^{-2s} = L^{2s} \sum_{n=1}^{\infty} x_n(b)^{-2s} \quad (15)$$

The L -dependence enters through both the explicit L^{2s} factor and the implicit $b = \lambda\beta = \lambda\hbar c/(L\bar{M}_{\text{Pl}}^2)$ dependence of the roots $x_n(b)$.

4.1.3 Explicit Form of $E_{\text{eff}}(L)$

Combining contributions and using the metrological anchor:

$$E_{\text{eff}}(L) = \frac{\hbar c}{L^3} + \frac{c_{\text{Cas}}}{L} \cdot F(b(L)) + \frac{\lambda\hbar c}{L^3 \bar{M}_5^3} \cdot G(b(L)) \quad (16)$$

where:

- First term: brane tension $\sigma = \hbar c/L^3$
- Second term: Casimir contribution with $c_{\text{Cas}} = -\pi^2/720$ (5D scalar)
- Third term: boundary coupling with $G(b)$ encoding mode contributions
- $F(b)$ and $G(b)$ are dimensionless functions of b

4.2 Stationarity Condition

Taking the derivative with respect to L :

$$\frac{dE_{\text{eff}}}{dL} = -\frac{3\hbar c}{L^4} - \frac{c_{\text{Cas}}}{L^2} F(b) + \frac{c_{\text{Cas}}}{L} F'(b) \frac{db}{dL} + \dots = 0 \quad (17)$$

Since $b = \lambda\hbar c/(L\bar{M}_{\text{Pl}}^2)$:

$$\frac{db}{dL} = -\frac{\lambda\hbar c}{L^2 \bar{M}_{\text{Pl}}^2} = -\frac{b}{L} \quad (18)$$

The stationarity condition becomes:

$$\boxed{3 + \frac{c_{\text{Cas}} L^3}{\hbar c} [F(b) + bF'(b)] + \lambda \cdot H(b) = 0} \quad (19)$$

where $H(b)$ collects boundary term contributions.

4.3 Analysis of Stationarity Equation

Lemma 4.1 (Asymptotic Limits). *The stationarity equation (19) has the following limits:*

1. **Neumann limit** ($b \rightarrow 0$): $F(0) = F_0$, $F'(0)$ finite. The equation reduces to:

$$L^3 = -\frac{3\hbar c}{c_{\text{Cas}} F_0} \quad (20)$$

2. **Large- b limit** ($b \rightarrow \infty$): $F(b) \sim b^{-\alpha}$ for some $\alpha > 0$. The equation becomes:

$$L^3 \sim \frac{3\hbar c}{c_{Cas}} b^\alpha \quad (21)$$

Proof. In the Neumann limit, the spectrum approaches $x_n \rightarrow n\pi$, giving standard Casimir results. In the large- b limit, the first mode approaches Dirichlet ($x_1 \rightarrow \pi/2$) while higher modes shift, suppressing the spectral sum. $\square$

4.4 Discrete Solutions with λ Quantization

Substituting $\lambda = |k|/(2\pi)$ into (19):

$$3 + \frac{c_{Cas} L^3}{\hbar c} [F(b_k) + b_k F'(b_k)] + \frac{|k|}{2\pi} H(b_k) = 0 \quad (22)$$

where $b_k = |k|\beta/(2\pi)$.

Proposition 4.2 (Discrete L_k Family). *For each $k \in \mathbb{Z}^+$, equation (22) admits a solution L_k satisfying:*

$$L_k = L_1 \cdot g(k) \quad (23)$$

where $g(k)$ is a monotonic function with $g(1) = 1$.

The explicit form of $g(k)$ depends on the functions F and H , which require detailed spectral calculations.

4.5 Route C Result

Main Result

Route C (Variational): The stationarity condition $dE_{\text{eff}}/dL = 0$ combined with λ -quantization yields a discrete family of solutions:

$$L_k = \frac{\hbar c}{\beta_k \cdot \bar{M}_{\text{Pl}}^2}, \quad k \in \mathbb{Z}^+ \quad (24)$$

where β_k satisfies the transcendental equation (22).

Tag: [D]+[P] for structure. [OPEN] for explicit β_k values (requires spectral function evaluation).

5 Route D: Spectral + Boundary Closure

5.1 Spectral Condition with Robin BC

From v26, the KK spectrum satisfies:

$$\tan(m_n L) = -\frac{m_b}{m_n} \quad (25)$$

Defining $x_n \equiv m_n L$ and $b \equiv m_b L$:

$$\tan(x_n) = -\frac{b}{x_n} \quad (26)$$

Tag: [D] from boundary variation.

5.2 λ Enters via Boundary Mapping

From v27, the brane-localized mass is:

$$m_b = \frac{\lambda\sigma}{M_5^3} \quad (27)$$

Using the bridge relation $\bar{M}_{\text{Pl}}^2 = M_5^3 L$:

$$m_b = \frac{\lambda\sigma L}{\bar{M}_{\text{Pl}}^2} \quad (28)$$

Therefore:

$$b = m_b L = \frac{\lambda\sigma L^2}{\bar{M}_{\text{Pl}}^2} = \lambda\beta \quad (29)$$

Tag: [D] from action variation and boundary matching.

5.3 Combined System

We now have a system of relations:

$$\text{Spectral: } \tan(x_1) = -\frac{b}{x_1} \quad (30)$$

$$\text{Robin parameter: } b = \lambda\beta \quad (31)$$

$$\text{Control: } \beta = \frac{\hbar c}{L\bar{M}_{\text{Pl}}^2} \quad (32)$$

$$\text{Quantization: } \lambda = \frac{|k|}{2\pi} \quad (33)$$

$$\text{Gap: } m_{\text{gap}} = \frac{x_1(b)}{L} \quad (34)$$

5.4 Self-Consistency Analysis

Lemma 5.1 (Parametric Family). *For fixed k and $\bar{M}_{\text{Pl}}$, the system (30)–(34) defines a one-parameter family of consistent solutions $(L, \beta, b, x_1, m_{\text{gap}})$ parameterized by L .*

Proof. Given L and k :

1. Compute $\beta = \hbar c / (L\bar{M}_{\text{Pl}}^2)$ from (32)
2. Compute $\lambda = |k| / (2\pi)$ from (33)
3. Compute $b = \lambda\beta$ from (31)
4. Solve $\tan(x_1) = -b/x_1$ for $x_1 \in (\pi/2, \pi)$ from (30)
5. Compute $m_{\text{gap}} = x_1/L$ from (34)

Each step is well-defined, proving consistency. □

5.5 Constraint from Spectral Bounds

The spectral equation (26) constrains x_1 :

$$\frac{\pi}{2} < x_1(b) < \pi \quad \text{for } b > 0 \quad (35)$$

Proposition 5.2 (Gap Bounds Without Identification). *The mass gap satisfies:*

$$\boxed{\frac{\pi}{2L} < m_{\text{gap}} < \frac{\pi}{L}} \quad (36)$$

Tag: [D] — no identification used.

5.6 Inverting for L

From (34):

$$L = \frac{x_1(b)}{m_{\text{gap}}} \quad (37)$$

But m_{gap} is unknown without identification!

Key insight: We can express L in terms of β without knowing m_{gap} explicitly:

$$L = \frac{\hbar c}{\beta \bar{M}_{Pl}^2} \quad (38)$$

The constraint becomes: what values of β are internally consistent?

5.7 β Consistency Equation

Substituting $L = \hbar c/(\beta \bar{M}_{Pl}^2)$ into $b = \lambda\beta$:

$$b = \frac{|k|}{2\pi}\beta \quad (39)$$

For each k , as β varies over $(0, \infty)$, b varies over $(0, \infty)$, and $x_1(b)$ traces the curve from π (Neumann) to $\pi/2$ (Dirichlet).

Proposition 5.3 (Continuous Family per k). *For each $k \in \mathbb{Z}^+$, there exists a continuous family of internally consistent solutions parameterized by $\beta \in (0, \infty)$:*

$$\mathcal{F}_k = \{(L(\beta), b(\beta), x_1(\beta), m_{\text{gap}}(\beta)) : \beta \in (0, \infty)\} \quad (40)$$

where:

$$L(\beta) = \frac{\hbar c}{\beta \bar{M}_{Pl}^2} \quad (41)$$

$$b(\beta) = \frac{|k|}{2\pi}\beta \quad (42)$$

$$x_1(\beta) = x_1\left(\frac{|k|}{2\pi}\beta\right) \quad (43)$$

$$m_{\text{gap}}(\beta) = \frac{x_1(\beta) \bar{M}_{Pl}^2}{\hbar c / \beta} = \frac{\beta x_1(\beta) \bar{M}_{Pl}^2}{\hbar c} \quad (44)$$

5.8 Route D Result

Main Result

Route D (Spectral): The spectral condition plus λ -quantization yields a discrete family of curves $\mathcal{F}_k$, one for each $k \in \mathbb{Z}^+$.

Each curve represents internally consistent $(L, \beta, m_{\text{gap}})$ triples. To select a *point* on a curve requires one additional constraint — either:

1. A variational principle (Route C) fixing β , or
2. An identification $m_{\text{gap}} = M_*$ [I]+[BL]

Tag: [D]+[P] for curve structure. [OPEN] for point selection.

6 Combined Analysis: Route C + Route D

6.1 Intersection of Constraints

Route C provides: β_k satisfying stationarity (22).

Route D provides: continuous family $\mathcal{F}_k$ for each k .

Theorem 6.1 (Discrete L_k Selection). *The intersection of Route C (variational) with Route D (spectral) selects discrete points (β_k^*, L_k^*) on each family $\mathcal{F}_k$:*

$$L_k^* = \frac{\hbar c}{\beta_k^* \bar{M}_{Pl}^2} \quad (45)$$

where β_k^* is the unique solution to (22) on $\mathcal{F}_k$.

6.2 Explicit Form for L_k

Combining the metrological anchor with the bridge relation:

$$\sigma L^3 = \hbar c \quad (46)$$

$$\bar{M}_{Pl}^2 = M_5^3 L \quad (47)$$

Eliminating σ :

$$L^2 = \frac{\hbar c}{\sigma L} = \frac{\hbar c \bar{M}_{Pl}^2}{\sigma L \bar{M}_{Pl}^2} = \frac{\hbar c}{\beta \bar{M}_{Pl}^2 \cdot L} \quad (48)$$

Thus:

$$L^3 = \frac{\hbar c}{\beta \bar{M}_{Pl}^2} \quad (49)$$

Wait, this contradicts (8). Let me recalculate.

From $\beta = \sigma L^2 / \bar{M}_{Pl}^2$ and $\sigma = \hbar c / L^3$:

$$\beta = \frac{\hbar c}{L^3} \cdot \frac{L^2}{\bar{M}_{Pl}^2} = \frac{\hbar c}{L \bar{M}_{Pl}^2} \quad (50)$$

So $L = \hbar c / (\beta \bar{M}_{Pl}^2)$, confirming (8). Good.

6.3 Numerical Scaling

In natural units ($\hbar = c = 1$):

$$L = \frac{1}{\beta \bar{M}_{Pl}^2} \quad (51)$$

With $\bar{M}_{Pl} = 2.435 \times 10^{18}$ GeV:

$$L = \frac{1}{\beta \times (5.93 \times 10^{36} \text{ GeV}^2)} \quad (52)$$

For small β , L is large (in GeV^{-1}).

Note: No identification was used — this table shows the internal $L(\beta)$ relationship.

Table 1: L vs β for various β values (no identification)

β	L (GeV $^{-1}$)	L (m)
10^{-30}	1.69×10^{-7}	3.33×10^{-23}
10^{-33}	1.69×10^{-4}	3.33×10^{-20}
10^{-36}	1.69×10^{-1}	3.33×10^{-17}
10^{-39}	1.69×10^2	3.33×10^{-14}

7 Constraint Window and Discretization

7.1 Bounds from Spectral Analysis

From the spectral bounds (35) and the gap relation:

$$\frac{\pi}{2L} < m_{\text{gap}} < \frac{\pi}{L} \quad (53)$$

This gives a constraint on (L, m_{gap}) pairs:

$$\frac{\pi}{2m_{\text{gap}}} < L < \frac{\pi}{m_{\text{gap}}} \quad (54)$$

7.2 β Window

Using $L = 1/(\beta \bar{M}_{\text{Pl}}^2)$ (natural units):

$$\frac{\pi m_{\text{gap}}}{2\bar{M}_{\text{Pl}}^2} < \beta < \frac{\pi m_{\text{gap}}}{\bar{M}_{\text{Pl}}^2} \quad (55)$$

This is a factor-of-2 window for β given any m_{gap} .

7.3 Discretization via λ

With $\lambda = |k|/(2\pi)$ and $b = \lambda\beta$:

$$b_k = \frac{|k|\beta}{2\pi} \quad (56)$$

Different k values give different b_k , hence different $x_1(b_k)$, hence different points on the consistency curve.

Proposition 7.1 (Discrete k -Branches). *The continuous family of consistent (L, β) pairs is organized into discrete k -branches:*

$$\mathcal{B}_k = \{(L, \beta) : b = |k|\beta/(2\pi), x_1 = x_1(b), m_{\text{gap}} = x_1\beta\bar{M}_{\text{Pl}}^2\} \quad (57)$$

Each branch is a curve in (L, β) space. Adjacent branches (k and $k+1$) are separated by a finite gap in b -space.

7.4 Weak Closure Statement

Closure Status

Weak Closure Achieved:

Without any identification:

1. L is constrained by internal consistency to the family $L = 1/(\beta \bar{M}_{\text{Pl}}^2)$
2. The λ -quantization organizes consistent solutions into discrete k -branches
3. Each k -branch $\mathcal{B}_k$ is a curve, not a point
4. Point selection on a branch requires one additional input

Gap Status: The derivation is [D]+[P]+[BL] for structure; [OPEN] for point selection without identification.

8 Compatibility with Earlier Results

8.1 Mapping to v27–v29 Symbols

Table 2: Symbol correspondence with v27–v29

This Note	v27–v29	Relation
L	$L = R_\xi$ (v22 convention)	Same
β	β (v29)	Same: $\sigma L^2 / \bar{M}_{\text{Pl}}^2$
b	$b = m_b L$ (v26)	Same: $\lambda \beta$
λ	λ (v27–v28)	Same: $ k /(2\pi)$
x_1	x_1 (v26)	Same: first root of $\tan x = -b/x$
σ	σ (v27)	Same: brane tension

8.2 Consistency Check: β from v29

In v29, β was computed using identification $L = \pi \hbar c / M_Z$:

$$\beta_{\text{v29}} = \frac{\hbar c}{L_{\text{v29}} \bar{M}_{\text{Pl}}^2} = \frac{M_Z}{\pi \bar{M}_{\text{Pl}}^2} \approx 4.89 \times 10^{-36} \quad (58)$$

This value lies on one of our k -branches. Which k ?

From $b = \lambda \beta = |k| \beta / (2\pi)$ and the Neumann-limit behavior ($b \ll 1 \Rightarrow x_1 \approx \pi$):

$$b_{k=1} = \frac{\beta}{2\pi} \approx \frac{4.89 \times 10^{-36}}{6.28} \approx 7.8 \times 10^{-37} \quad (59)$$

This extremely small b confirms the Neumann limit. All reasonable k values give $b \ll 1$.

8.3 Consistency Box

Consistency Verification

1. β definition: ✓ Same as v29 (6)
2. $b = \lambda\beta$: ✓ Same as v27 (29)
3. Spectral equation: ✓ Same as v26 (26)
4. λ quantization: ✓ Same as v28 (10)
5. Metrological anchor: ✓ Same as v27 (3)

All relations are internally consistent with v26–v29.

9 Results Summary

9.1 Main Results

Theorem 9.1 (Interval Constraint Without Identification). *The interval length L satisfies:*

$$L = \frac{\hbar c}{\beta \bar{M}_{Pl}^2} \quad (60)$$

where β is constrained to lie on a discrete k -branch $\mathcal{B}_k$ for some $k \in \mathbb{Z}^+$.

The k -branches are related by:

$$b_k = \frac{|k|}{2\pi} \beta \quad (61)$$

Tags: $[D]$ for main relation, $[P]$ for k -quantization, $[BL]$ for $\bar{M}_{Pl}$.

9.2 Discrete vs Continuous

Table 3: Closure status summary

Quantity	Status	Tag
$L = f(\beta, \bar{M}_{Pl})$	Derived	$[D]+[BL]$
β on k -branch	Constrained	$[D]+[P]$
Specific β value	Open	$[OPEN]$
Specific L value	Open	$[OPEN]$
k -branch selection	Open	$[OPEN]$

9.3 What Would Close L Fully

To achieve strong closure, one needs:

1. **Option A:** A variational principle that uniquely determines β^*
2. **Option B:** A topological argument that selects a specific k
3. **Option C:** An identification $m_{\text{gap}} = M_*$ $[I]+[BL]$ (the current approach)

Options A and B remain $[OPEN]$. Option C is what v21–v29 use.

10 Epistemic Ledger

Table 4: Epistemic status of all quantities

Quantity	Tag	Notes
$\hbar, c$	[BL]	SI 2019 exact
$\bar{M}_{\text{Pl}}$	[BL]	2.435×10^{18} GeV
$\sigma L^3 = \hbar c$	[D]	Metrological anchor
$\bar{M}_{\text{Pl}}^2 = M_5^3 L$	[D]	Bridge relation
$\beta = \sigma L^2 / \bar{M}_{\text{Pl}}^2$	[D]	Control parameter
$\tan(x) = -b/x$	[D]	Spectral equation
$b = \lambda\beta$	[D]	Robin parameter
$\lambda = k /(2\pi)$	[P]	Topological quantization
$L = \hbar c / (\beta \bar{M}_{\text{Pl}}^2)$	[D]+[BL]	Main result
β value	[OPEN]	Not fixed without identification
L value	[OPEN]	Not fixed without identification
k selection	[OPEN]	Not fixed

11 Reviewer Trap Checklist

Table 5: Reviewer trap verification

#	Trap	Check	Status
1	Hidden M_Z via numerics	No 91.19 GeV appears	PASS
2	Sneaking ℓ_P or G_N	Neither used as input	PASS
3	Boundary term not derived	Eq. (27) from action	PASS
4	Missing SA justification	v28 Track A referenced	PASS
5	L vs $R = L/\pi$ confusion	Convention stated: $L = \text{interval}$	PASS
6	Build artifacts committed	Verified not staged	PASS
7	Circular β definition	β defined, not assumed	PASS
8	Dimension errors	Appendix A verification	PASS
9	$\bar{M}_{\text{Pl}}$ vs M_{Pl}	Reduced Planck used, stated	PASS
10	Route C functional ad-hoc	Constructed from action	PASS
11	Route D doesn't use λ	λ enters via $b = \lambda\beta$	PASS
12	No discrete output	k -branches derived	PASS
13	Missing uncertainty	No numeric values claimed	N/A
14	Claim strong closure falsely	Weak closure stated	PASS

12 Conclusions

1. The interval length L can be expressed as $L = \hbar c / (\beta \bar{M}_{\text{Pl}}^2)$ using only derived relations and baseline constants [D]+[BL].
2. The λ -quantization $\lambda = |k|/(2\pi)$ organizes consistent solutions into discrete k -branches [P].

3. Each k -branch is a continuous curve in (L, β) space; point selection requires additional input [OPEN].
4. Strong closure (unique L value) is NOT achieved without identification.
5. Weak closure IS achieved: L is constrained to a discrete family of curves, not an arbitrary value.
6. The identification $m_{\text{gap}} = M_Z$ [I]+[BL] selects a specific point on a k -branch, giving $\beta \approx 5 \times 10^{-36}$ and $L \approx 3.4 \times 10^{-2} \text{ GeV}^{-1}$.

A Dimensional and Convention Audit

A.1 SI Dimensions

Table 6: Dimensional analysis in SI units

Quantity	SI Dimension	Natural Units
$\hbar$	$\text{kg} \cdot \text{m}^2 \cdot \text{s}^{-1}$	1
c	$\text{m} \cdot \text{s}^{-1}$	1
$\bar{M}_{\text{Pl}}$	kg	[M]
σ	$\text{kg} \cdot \text{m}^{-3}$	$[\text{M}]^4$
L	m	$[\text{M}]^{-1}$
M_5	kg	[M]
β	1	1
b	1	1
λ	1	1
m_b	m^{-1}	[M]
m_{gap}	m^{-1}	[M]

A.2 Dimensional Verification

Check 1: $[\beta] = 1$

$$[\beta] = \frac{[\sigma][L]^2}{[\bar{M}_{\text{Pl}}]^2} = \frac{[M]^4 \cdot [M]^{-2}}{[M]^2} = 1 \quad \checkmark \quad (62)$$

Check 2: $[b] = 1$

$$[b] = [m_b][L] = [M] \cdot [M]^{-1} = 1 \quad \checkmark \quad (63)$$

Check 3: Metrological anchor

$$[\sigma L^3] = [M]^4 \cdot [M]^{-3} = [M] = [\hbar c] \quad \checkmark \quad (64)$$

Check 4: Bridge relation

$$[\bar{M}_{\text{Pl}}^2] = [M]^2, \quad [M_5^3 L] = [M]^3 \cdot [M]^{-1} = [M]^2 \quad \checkmark \quad (65)$$

Check 5: Main result

$$\left[\frac{\hbar c}{\beta \bar{M}_{\text{Pl}}^2} \right] = \frac{[M]}{1 \cdot [M]^2} = [M]^{-1} = [L] \quad \checkmark \quad (66)$$

A.3 Planck Mass Convention Map

$$M_{\text{Pl}} = \sqrt{8\pi} \bar{M}_{\text{Pl}} \approx 5.01 \bar{M}_{\text{Pl}} \quad (67)$$

$$\bar{M}_{\text{Pl}} = 2.435 \times 10^{18} \text{ GeV} \quad (68)$$

$$M_{\text{Pl}} = 1.221 \times 10^{19} \text{ GeV} \quad (69)$$

A.4 Length Convention Map

$$L = \text{interval length (used throughout)} \quad (70)$$

$$R = L/\pi = \text{“radius” in circle picture} \quad (71)$$

$$R_\xi = L = \text{v22 canonical convention} \quad (72)$$

A.5 β Under Convention Change

If using R instead of L :

$$\beta_R = \frac{\sigma R^2}{\bar{M}_{\text{Pl}}^2} = \frac{\sigma(L/\pi)^2}{\bar{M}_{\text{Pl}}^2} = \frac{\beta_L}{\pi^2} \quad (73)$$

If using M_{Pl} instead of $\bar{M}_{\text{Pl}}$:

$$\beta^{(\text{orig})} = \frac{\sigma L^2}{M_{\text{Pl}}^2} = \frac{\beta^{(\text{red})}}{8\pi} \quad (74)$$

B Numerical Implementation Details

B.1 Root Finding for $x_1(b)$

The spectral equation $\tan(x) = -b/x$ for $x \in (\pi/2, \pi)$ is solved using Brent’s method with:

- Bracket: $[(\pi/2)(1 + \epsilon), \pi(1 - \epsilon)]$ with $\epsilon = 10^{-10}$
- Tolerance: 10^{-12}
- Residual check: $|\tan(x_1) + b/x_1| < 10^{-10}$

B.2 Asymptotic Expansions

For small b (Neumann limit):

$$x_1(b) = \pi - b + \frac{b^2}{\pi} - \frac{b^3}{\pi^2} + O(b^4) \quad (75)$$

For large b (Dirichlet limit):

$$x_1(b) = \frac{\pi}{2} + \frac{\pi}{2b} + O(b^{-2}) \quad (76)$$

B.3 Asymptotic Expansion Details

The small- b expansion coefficients arise from perturbation theory around the Neumann solution $x = \pi$. Let $x = \pi - \delta$ with $\delta \ll 1$:

$$\tan(\pi - \delta) = \tan(-\delta) = -\delta - \frac{\delta^3}{3} + O(\delta^5) \quad (77)$$

Substituting into $\tan(x) = -b/x$:

$$-\delta - \frac{\delta^3}{3} = -\frac{b}{\pi - \delta} \quad (78)$$

Expanding the RHS:

$$-\frac{b}{\pi - \delta} = -\frac{b}{\pi} \left(1 + \frac{\delta}{\pi} + \frac{\delta^2}{\pi^2} + \dots \right) \quad (79)$$

At leading order: $\delta = b/\pi$. At next order:

$$\delta = b/\pi + b^2/\pi^2 + O(b^3) \quad (80)$$

Therefore:

$$x_1 = \pi - b/\pi - b^2/\pi^2 + O(b^3) \approx \pi - b + O(b^2) \quad (81)$$

The large- b expansion uses $x \rightarrow \pi/2$ as the leading approximation:

$$\tan(x) \rightarrow \infty \quad \text{as} \quad x \rightarrow \pi/2^+ \quad (82)$$

For $b \gg 1$, let $x = \pi/2 + \epsilon$ with $\epsilon \ll 1$:

$$\tan(\pi/2 + \epsilon) = -\cot(\epsilon) \approx -1/\epsilon \quad (83)$$

Thus:

$$-\frac{1}{\epsilon} = -\frac{b}{\pi/2 + \epsilon} \approx -\frac{2b}{\pi} \quad (84)$$

Solving: $\epsilon = \pi/(2b)$, giving:

$$x_1 = \frac{\pi}{2} + \frac{\pi}{2b} + O(b^{-2}) \quad (85)$$

B.4 Consistency Tables

Table 7: L vs β consistency (natural units, $\hbar = c = 1$)

β	L (GeV ⁻¹)	$b_{k=1}$	$x_1(b_{k=1})$
10^{-34}	1.686×10^{-3}	1.592×10^{-35}	$\pi - 1.6 \times 10^{-35}$
10^{-35}	1.686×10^{-2}	1.592×10^{-36}	$\pi - 1.6 \times 10^{-36}$
10^{-36}	1.686×10^{-1}	1.592×10^{-37}	$\pi - 1.6 \times 10^{-37}$
10^{-37}	1.686×10^0	1.592×10^{-38}	$\pi - 1.6 \times 10^{-38}$

All entries satisfy the internal consistency relations exactly.

C Extended Derivations

C.1 Derivation of M_5 from Bridge Relation

From the bridge relation $\bar{M}_{\text{Pl}}^2 = M_5^3 L$:

$$M_5^3 = \frac{\bar{M}_{\text{Pl}}^2}{L} \quad (86)$$

Taking the cube root:

$$M_5 = \left(\frac{\bar{M}_{\text{Pl}}^2}{L} \right)^{1/3} = \bar{M}_{\text{Pl}}^{2/3} L^{-1/3} \quad (87)$$

Using $L = \hbar c / (\beta \bar{M}_{\text{Pl}}^2)$:

$$M_5 = \bar{M}_{\text{Pl}}^{2/3} \left(\frac{\beta \bar{M}_{\text{Pl}}^2}{\hbar c} \right)^{1/3} \quad (88)$$

Simplifying:

$$M_5 = \bar{M}_{\text{Pl}}^{2/3} \cdot \beta^{1/3} \cdot \bar{M}_{\text{Pl}}^{2/3} \cdot (\hbar c)^{-1/3} = \bar{M}_{\text{Pl}}^{4/3} \beta^{1/3} (\hbar c)^{-1/3} \quad (89)$$

C.2 Derivation of σ from L

From the metrological anchor $\sigma L^3 = \hbar c$:

$$\sigma = \frac{\hbar c}{L^3} \quad (90)$$

Using $L = \hbar c / (\beta \bar{M}_{\text{Pl}}^2)$:

$$\sigma = \frac{\hbar c}{\left(\frac{\hbar c}{\beta \bar{M}_{\text{Pl}}^2} \right)^3} = \frac{(\hbar c) \cdot (\beta \bar{M}_{\text{Pl}}^2)^3}{(\hbar c)^3} \quad (91)$$

Simplifying:

$$\sigma = \frac{\beta^3 \bar{M}_{\text{Pl}}^6}{(\hbar c)^2} \quad (92)$$

In natural units ($\hbar = c = 1$):

$$\sigma = \beta^3 \bar{M}_{\text{Pl}}^6 \quad (93)$$

C.3 Consistency Check: β Definition

Verify $\beta = \sigma L^2 / \bar{M}_{\text{Pl}}^2$ using (92):

$$\beta = \frac{\sigma L^2}{\bar{M}_{\text{Pl}}^2} = \frac{\beta^3 \bar{M}_{\text{Pl}}^6 \cdot L^2}{\bar{M}_{\text{Pl}}^2} = \beta^3 \bar{M}_{\text{Pl}}^4 L^2 \quad (94)$$

Using $L = 1 / (\beta \bar{M}_{\text{Pl}}^2)$:

$$\beta = \beta^3 \bar{M}_{\text{Pl}}^4 \cdot \frac{1}{\beta^2 \bar{M}_{\text{Pl}}^4} = \beta \quad \checkmark \quad (95)$$

C.4 Robin Parameter Verification

The Robin parameter $b = \lambda \beta$. Using $m_b = \lambda \sigma / M_5^3$:

$$b = m_b L = \frac{\lambda \sigma L}{M_5^3} \quad (96)$$

From bridge relation: $M_5^3 = \bar{M}_{\text{Pl}}^2 / L$, so:

$$b = \frac{\lambda \sigma L}{\bar{M}_{\text{Pl}}^2 / L} = \frac{\lambda \sigma L^2}{\bar{M}_{\text{Pl}}^2} = \lambda \beta \quad \checkmark \quad (97)$$

C.5 Gap Expression in Terms of β

From $m_{\text{gap}} = x_1 / L$ and $L = 1 / (\beta \bar{M}_{\text{Pl}}^2)$:

$$m_{\text{gap}} = x_1 \cdot \beta \bar{M}_{\text{Pl}}^2 \quad (98)$$

The bounds $\pi/2 < x_1 < \pi$ translate to:

$$\frac{\pi}{2} \beta \bar{M}_{\text{Pl}}^2 < m_{\text{gap}} < \pi \beta \bar{M}_{\text{Pl}}^2 \quad (99)$$

C.6 k -Dependence of x_1

For topological charge k , we have $b_k = |k|\beta/(2\pi)$. The spectral root $x_1(b_k)$ depends implicitly on k :

$$x_1^{(k)} \equiv x_1 \left(\frac{|k|\beta}{2\pi} \right) \quad (100)$$

For small β (Neumann limit), using the expansion (75):

$$x_1^{(k)} \approx \pi - \frac{|k|\beta}{2\pi} \quad (101)$$

The gap for each k -branch:

$$m_{\text{gap}}^{(k)} = x_1^{(k)} \cdot \beta \bar{M}_{\text{P1}}^2 \approx \left(\pi - \frac{|k|\beta}{2\pi} \right) \beta \bar{M}_{\text{P1}}^2 \quad (102)$$

Expanding:

$$m_{\text{gap}}^{(k)} \approx \pi \beta \bar{M}_{\text{P1}}^2 - \frac{|k|\beta^2 \bar{M}_{\text{P1}}^2}{2\pi} \quad (103)$$